

Experiment No. 3

Creating a sample database (Library Database System)

using any RDBMS package e.g. MySQL.

Use of DDL commands/queries

```
create database Library_DB_SE_2025_26;
```

```
use Library_DB_SE_2025_26;
```

```
select database();
```

Create a table - Publisher

```
create table Publisher (name varchar (40) ,
```

```
                        address varchar (30) ,
```

```
                        phone int);
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL > show
tables;
```

```
+-----+
| | Tables_in_library_db_se_2025_26 |
+-----+
| publisher |
+-----+
```

```
1 row in set (0.0014 sec)
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL >desc
publisher;
```

```
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| name | varchar(40) | YES | | NULL | |
| address | varchar(30) | YES | | NULL | |
| phone | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
```

```
3 rows in set (0.0015 sec)
```

Add primary key constraint

```
alter table Publisher
```

```
add primary key (name);
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL > desc
publisher;
```



```
#Add Referential Integrity constraint (i.e. Foreign key
constraint)
alter table Book
add foreign key (publisher_name) references publisher (name) on
delete cascade;
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL > desc book;
```

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	
title	varchar(40)	YES		NULL	
pub_year	varchar(20)	YES		NULL	
publisher_name	varchar(40)	YES	MUL	NULL	

4 rows in set (0.0014 sec)

```
# Create a table - Book_Authors
create table Book_Authors (book_id int,
                           author_name varchar (30),
                           primary key (book_id, author_name),
                           foreign key (book_id) references book (book_id) on
delete cascade);
```

```
# Create a table - Library_Branch
create table Library_Branch (branch_id int primary key,
                             branch_name varchar (40),
                             address varchar (30));
```

```
#Add unique constraint
```

```
alter table Library_Branch
change branch_name branch_name varchar (40) not null unique;
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL > desc
library_branch;
```

Field	Type	Null	Key	Default	Extra
branch_id	int	NO	PRI	NULL	
branch_name	varchar(40)	NO	UNI	NULL	
address	varchar(30)	YES		NULL	

3 rows in set (0.0015 sec)

```
# Create a table - Book_Copies
create table Book_Copies      (book_id int,
                              branch_id int,
                              no_of_copies int,
                              primary key (book_id, branch_id),
foreign key (book_id) references book (book_id) on delete cascade,
foreign key (branch_id) references library_branch (branch_id) on
delete cascade);
```

```
#Add check constraint
alter table Book_Copies
add check (no_of_copies>0);
```

```
# Create the table - Borrower
create table Borrower
      (card_no int primary key,
       name varchar (30) not null,
       address varchar (50),
       phone bigint);
```

```
# Create the table - Book_Loans
create table Book_Loans
      (book_id int,
       branch_id int,
       card_no int not null,
       date_out date,
       due_date date,
       primary key (book_id, branch_id, card_no),
foreign key (book_id) references book (book_id) on delete cascade,
foreign key (branch_id) references library_branch (branch_id) on
delete cascade,
foreign key(card_no) references Borrower (card_no) on delete
cascade);
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL >show tables;
```

```
+-----+
| Tables_in_library_db_se_2025_26 |
+-----+
| book |
| book_authors |
| book_copies |
| book_loans |
| borrower |
| library_branch |
| publisher |
+-----+
```

```
7 rows in set (0.0011 sec)
```

```
# To rename a table
```

```
rename table Borrower to Borrower1;
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL >show tables;
```

```
+-----+
| Tables_in_library_db_se_2025_26 |
+-----+
| book                               |
| book_authors                       |
| book_copies                        |
| book_loans                         |
| borrower1                          |
| library_branch                     |
| publisher                          |
+-----+
```

```
7 rows in set (0.0013 sec)
```

```
rename table Borrower1 to Borrower;
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL >show tables;
```

```
+-----+
| Tables_in_library_db_se_2025_26 |
+-----+
| book                               |
| book_authors                       |
| book_copies                        |
| book_loans                         |
| borrower                           |
| library_branch                     |
| publisher                          |
+-----+
```

```
7 rows in set (0.0049 sec)
```

```
#drop database Library_DB_SE_2025_26;
```

```
#drop table publisher;
```

Experiment No. 4

SQL DML queries: Use of SQL DML queries to retrieve the data,

insert the data, delete the record and update the database.

```
use Library_DB_SE_2025_26;
select database();
```

To insert records into Publisher table

```
insert into Publisher values ('Mcgraw-Hill', 'Bangalore', 9989076587);
insert into Publisher values ('Pearson', 'NewDelhi', 9889076565);
insert into Publisher values ('Random House', 'Hyderabad', 7455679345);
insert into Publisher values ('Hachette Livre', 'Chennai', 8970862340);
insert into Publisher values ('Grupo Planeta', 'Bangalore', 7756120238);
insert into Publisher values ('Wiley India', 'New Delhi', 1143630000);
insert into Publisher values ('S Chand', 'Noida', 7291975264);
insert into Publisher values ('Technical Publications', 'Pune', 9763719464);
insert into Publisher values ('Nirali Prakashan', 'Pune', 2025512336);
insert into Publisher values ('Prakash Publication', 'Jalgaon', 9272514482);
insert into Publisher values ('Galgotia Publications', 'New Delhi', 1123263334);
```

MySQL localhost:3306 ssl library_db_se_2025_26 SQL > select * from publisher;

name	address	phone
Galgotia Publications	New Delhi	1123263334
Grupo Planeta	Bangalore	7756120238
Hachette Livre	Chennai	8970862340
Mcgraw-Hill	Bangalore	9989076587
Nirali Prakashan	Pune	2025512336
Pearson	New Delhi	9889076565
Prakash Publication	Jalgaon	9272514482
Random House	Hyderabad	7455679345
S Chand	Noida	7291975264
Technical Publications	Pune	9763719464
Wiley India	New Delhi	1143630000

11 rows in set (0.0115 sec)

To insert records into Book table

```
insert into book values (1,'DBMS','2017', 'Mcgraw-Hill');
insert into book values (2,'ADBMS','2016', 'Mcgraw-Hill');
insert into book values (3,'CN','2016', 'Pearson');
insert into book values (4,'CG','2015', 'Grupo Planeta');
insert into book values (5,'OS','2016', 'Pearson');
insert into book values (6,'CO','2008', 'Nirali Prakashan');
insert into book values (7,'CO','2005', 'Galgotia Publications');
insert into book values (8,'DBMS','2022', 'Technical Publications');
insert into book values (9,'DBMS for Distributed System','2022', 'Pearson');
```

```
MySQL localhost:3306 ssl library_db_se 2025_26 SQL > select * from Book;
```

book_id	title	pub_year	publisher_name
1	DBMS	2017	Mcgraw-Hill
2	ADBMS	2016	Mcgraw-Hill
3	CN	2016	Pearson
4	CG	2015	Grupo Planeta
5	OS	2016	Pearson
6	CO	2008	Nirali Prakashan
7	CO	2005	Galgotia Publications
8	DBMS	2022	Technical Publications
9	DBMS for Distributed System	2022	Pearson

```
9 rows in set (0.0006 sec)
```

To insert records into Book_Authors table

```
insert into book_authors values (1, 'Navathe');
insert into book_authors values (2, 'Navathe');
insert into book_authors values (3, 'Tanenbaum');
insert into book_authors values (4, 'Edward Angel');
insert into book_authors values (5, 'Galvin');
insert into book_authors values (6, 'K. P. Adhiya');
insert into book_authors values (7, 'Navin Kumar');
insert into book_authors values (8, 'A. A. Puntambekar');
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL > select * from Book_Authors;
```

book_id	author_name
1	Navathe
2	Navathe
3	Tanenbaum
4	Edward Angel
5	Galvin
6	K. P. Adhiya
7	Navin Kumar
8	A. A. Puntambekar

```
8 rows in set (0.0008 sec)
```

To insert records into Library_Branch table

```
insert into library_branch values (10, 'RR Nagar', 'Bangalore');
insert into library_branch values (11, 'RNSIT', 'Bangalore');
insert into library_branch values (12, 'Rajaji Nagar', 'Bangalore');
insert into library_branch values (13, 'NITTE', 'Mangalore');
insert into library_branch values (14, 'Manipal', 'Udupi');
insert into library_branch values (15, 'SSBT', 'Jalgaon');
insert into library_branch values (16, 'SPIT', 'Mumbai');
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL > select *
from Library_Branch;
```

branch_id	branch_name	address
10	RR Nagar	Bangalore
11	RNSIT	Bangalore
12	Rajaji Nagar	Bangalore
13	NITTE	Mangalore
14	Manipal	Udupi
15	SSBT	Jalgaon
16	SPIT	Mumbai

```
7 rows in set (0.0008 sec)
```

```
# To insert records into Book_Copies table
```

```
insert into book_copies values (1, 10, 10);
insert into book_copies values (1, 11, 5);
insert into book_copies values (2, 12, 2);
insert into book_copies values (2, 13, 5);
insert into book_copies values (3, 14, 7);
insert into book_copies values (5, 10, 1);
insert into book_copies values (4, 11, 3);
insert into book_copies values (6, 15, 13);
insert into book_copies values (6, 16, 10);
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL > select * from
Book_Copies;
```

book_id	branch_id	no_of_copies
1	10	10
1	11	5
2	12	2
2	13	5
3	14	7
4	11	3
5	10	1
6	15	13
6	16	10

```
9 rows in set (0.0008 sec)
```

```
# To insert records into Borrower table
```

```
insert into borrower values (100, 'George', 'Jalgaon', 1122334455 );
insert into borrower values (101, 'Krishna', 'Daryapur', 9421516499 );
insert into borrower values (102, 'Uday', 'Dhule', 3322445566 );
insert into borrower values (103, 'Nilesh', 'Nagpur', 7722445566 );
insert into borrower values (104, 'Abhijit', 'Amravati', 7777445566 );
insert into borrower values (105, 'Hitesh', 'Amravati', 8888445566 );
```



```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL > select *
from borrower;
```

card_no	name	address	phone
100	George	Jalgaon	1122334455
101	Krishna	Daryapur	9421516499
102	Uday	Dhule	3322445566
103	Nilesh	Nagpur	7722445566
104	Abhijit	Amravati	7777445566
105	Hitesh	Amravati	8888445566

```
6 rows in set (0.0006 sec)
```

```
# To insert records into Book_Loans table
```

```
insert into book_loans values (1, 10, 101, '2017-01-10', '2017-06-17');
insert into book_loans values (3, 14, 101, '2017-03-18', '2017-07-17');
insert into book_loans values (2, 13, 101, '2017-02-11', '2017-04-21');
insert into book_loans values (4, 11, 101, '2017-03-15', '2017-07-15');
insert into book_loans values (1, 11, 104, '2017-04-12', '2017-05-12');
```

```
MySQL localhost:3306 ssl library_db_se_2025_26 SQL > select *
from Book_Loans;
```

book_id	branch_id	card_no	date_out	due_date
1	10	101	2017-01-10	2017-06-17
1	11	104	2017-04-12	2017-05-12
2	13	101	2017-02-11	2017-04-21
3	14	101	2017-03-18	2017-07-17
4	11	101	2017-03-15	2017-07-15

```
5 rows in set (0.0006 sec)
```

```
# List the names of all publishers
```

```
select name
from Publisher;
```

name
Galgotia Publications
Grupo Planeta
Hachette Livre
Mcgraw-Hill
Nirali Prakashan
Pearson

```

| Prakash Publication |
| Random House       |
| S Chand             |
| Technical Publications |
| Wiley India         |
+-----+
11 rows in set (0.0023 sec)

```

```

# List the names of all publishers with contact number
select name, phone
from Publisher;

```

```

+-----+-----+
| name                | phone      |
+-----+-----+
| Galgotia Publications | 1123263334 |
| Grupo Planeta         | 7756120238 |
| Hachette Livre        | 8970862340 |
| Mcgraw-Hill           | 9989076587 |
| Nirali Prakashan      | 2025512336 |
| Pearson               | 9889076565 |
| Prakash Publication   | 9272514482 |
| Random House          | 7455679345 |
| S Chand               | 7291975264 |
| Technical Publications | 9763719464 |
| Wiley India           | 1143630000 |
+-----+-----+
11 rows in set (0.0007 sec)

```

```

# List names of all publishers from New Delhi
select name
from Publisher
where address = 'New Delhi';

```

```

+-----+
| name                |
+-----+
| Galgotia Publications |
| Wiley India           |
+-----+
3 rows in set (0.0007 sec)

```

```

#Use of update
update Publisher
set address = 'New Delhi'
where name = 'Pearson';

```

```
select name
from Publisher
where address = 'New Delhi';
```

```
+-----+
| name          |
+-----+
| Galgotia Publications |
| Pearson        |
| Wiley India    |
+-----+
3 rows in set (0.0007 sec)
```

```
# List the titles of books published by publishers
select title
from Book;
```

```
+-----+
| title          |
+-----+
| DBMS           |
| ADBMS          |
| CN             |
| CG             |
| OS             |
| CO             |
| CO             |
| DBMS           |
| DBMS for Distributed System |
+-----+
9 rows in set (0.0022 sec)
```

```
select distinct title
from Book;
```

```
+-----+
| title          |
+-----+
| DBMS           |
| ADBMS          |
| CN             |
| CG             |
| OS             |
| CO             |
| DBMS for Distributed System |
+-----+
7 rows in set (0.0006 sec)
```

Display the details of books, alphabetically titled

```
select *  
from Book  
order by title;
```

book_id	title	pub_year	publisher_name
2	ADBMS	2016	Mcgraw-Hill
4	CG	2015	Grupo Planeta
3	CN	2016	Pearson
6	CO	2008	Nirali Prakashan
7	CO	2005	Galgotia Publications
1	DBMS	2017	Mcgraw-Hill
8	DBMS	2022	Technical Publications
9	DBMS for Distributed System	2022	Pearson
5	OS	2016	Pearson

9 rows in set (0.0007 sec)

```
select title  
from Book  
order by title desc;
```

title
OS
DBMS for Distributed System
DBMS
DBMS
CO
CO
CN
CG
ADBMS

9 rows in set (0.0007 sec)

#Find the latest publication year for DBMS book

```
select title, pub_year  
from Book  
where title = 'DBMS'  
order by pub_year desc  
limit 1;
```

title	pub_year
DBMS	2022

1 row in set (0.0104 sec)

```
# Display the details of books, publication year wise
select *
from Book
order by pub_year;
```

```
select title, pub_year
from Book
order by pub_year desc;
```

```
+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                  | 2022     |
| DBMS for Distributed System | 2022     |
| DBMS                  | 2017     |
| ADBMS                 | 2016     |
| CN                    | 2016     |
| OS                    | 2016     |
| CG                    | 2015     |
| CO                    | 2008     |
| CO                    | 2005     |
+-----+-----+
9 rows in set (0.0007 sec)
```

```
# To delete records
select * from book_copies;
delete from book_copies
where (book_id=1 and branch_id=10)
or (book_id=1 and branch_id=11)
or (book_id=2 and branch_id=12);
select * from book_copies;
```

```
insert into book_copies values (1, 10, 10);
insert into book_copies values (1, 11, 5);
insert into book_copies values (2, 12, 2);
select * from book_copies;
```

```
# List title of books and publication year, published between 2015
to 2022. Display recent publishing year on top.
```

```
select title, pub_year
from Book
where pub_year between '2015' and '2022'
order by pub_year desc;
```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                 | 2022     |
| DBMS for Distributed System | 2022     |
| DBMS                 | 2017     |
| ADBMS                | 2016     |
| CN                   | 2016     |
| OS                   | 2016     |
| CG                   | 2015     |
+-----+-----+

```

7 rows in set (0.0005 sec)

List title of books and publication year, published between 2017 to 2022.

```

select title, pub_year
from Book
where pub_year between '2017' and '2022';

```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                 | 2017     |
| DBMS                 | 2022     |
| DBMS for Distributed System | 2022     |
+-----+-----+

```

3 rows in set (0.0006 sec)

Find out average number of copies of each book

```

select book_id, avg(no_of_copies) as Average_No_of_Copies_Per_Book
from Book_Copies
group by book_id;

```

```

+-----+-----+
| book_id | Average_No_of_Copies_Per_Book |
+-----+-----+
| 1       | 7.5000                        |
| 2       | 3.5000                        |
| 3       | 7.0000                        |
| 4       | 3.0000                        |
| 5       | 1.0000                        |
| 6       | 11.5000                       |
+-----+-----+

```

6 rows in set (0.0010 sec)

Display the details of books along with average number of copies of each book

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id, author_name, title;
```

```
+-----+-----+-----+-----+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
|      1 | Navathe    | DBMS  |              7.5000 |
|      2 | Navathe    | ADBMS |              3.5000 |
|      3 | Tanenbaum  | CN     |              7.0000 |
|      4 | Edward Angel | CG     |              3.0000 |
|      5 | Galvin     | OS     |              1.0000 |
|      6 | K. P. Adhiya | CO     |             11.5000 |
+-----+-----+-----+-----+
6 rows in set (0.0010 sec)
```

Find out Authors whose average no of book copies in library are more than 3 and display the name of author on top having maximum average number of book copies.

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id,author_name, title
having Average_No_of_Copies_Per_Book >3
order by Average_No_of_Copies_Per_Book desc;
```

```
+-----+-----+-----+-----+
-+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
-+
|      6 | K. P. Adhiya | CO     |             11.5000 |
|      1 | Navathe    | DBMS  |              7.5000 |
|      3 | Tanenbaum  | CN     |              7.0000 |
|      2 | Navathe    | ADBMS |              3.5000 |
+-----+-----+-----+-----+
-+
4 rows in set (0.0008 sec)
```

```
# Display - title, publication year and publisher name for all titles
which includes DBMS.
```

```
select title, pub_year, publisher_name
from Book
where title like '%DBMS%';
```

```
+-----+-----+-----+
| title                | pub_year | publisher_name      |
+-----+-----+-----+
| DBMS                 | 2017     | McGraw-Hill         |
| ADBMS                | 2016     | McGraw-Hill         |
| DBMS                 | 2022     | Technical Publications |
| DBMS for Distributed System | 2022     | Pearson              |
+-----+-----+-----+
4 rows in set (0.0006 sec)
```

```
# Display the details of publishers from Pune and Jalgaon.
```

```
select *
from publisher
where address = 'Pune' or address = 'Jalgaon';
```

```
select *
from publisher
where address in ('Pune', 'Jalgaon');
```

```
+-----+-----+-----+
| name                | address  | phone              |
+-----+-----+-----+
| Nirali Prakashan    | Pune     | 2025512336        |
| Prakash Publication | Jalgaon  | 9272514482        |
| Technical Publications | Pune     | 9763719464        |
+-----+-----+-----+
3 rows in set (0.0045 sec)
```

```
#Display the details of publishers other than Pune and Jalgaon.
```

```
select *
from publisher
where address not in ('Pune', 'Jalgaon');
```


name	address	phone
Galgotia Publications	New Delhi	1123263334
Grupo Planeta	Bangalore	7756120238
Hachette Livre	Chennai	8970862340
Mcgraw-Hill	Bangalore	9989076587
Pearson	New Delhi	9889076565
Random House	Hyderabad	7455679345
S Chand	Noida	7291975264
Wiley India	New Delhi	1143630000

8 rows in set (0.0006 sec)

Use of IS NULL, IS NOT NULL

```
select *
from publisher
where phone is null;
```

```
select *
from publisher
where phone is not null;
```

name	address	phone
Galgotia Publications	New Delhi	1123263334
Grupo Planeta	Bangalore	7756120238
Hachette Livre	Chennai	8970862340
Mcgraw-Hill	Bangalore	9989076587
Nirali Prakashan	Pune	2025512336
Pearson	New Delhi	9889076565
Prakash Publication	Jalgaon	9272514482
Random House	Hyderabad	7455679345
S Chand	Noida	7291975264
Technical Publications	Pune	9763719464
Wiley India	New Delhi	1143630000

11 rows in set (0.0007 sec)

Experiment No. 5

To use various MySQL Commands/Queries such as - use of Aggregate Functions e.g. SUM(), COUNT(), AVG() etc., GROUP BY clause, HAVING clause, ORDER BY clause etc.

```
use Library_DB_SE_2025_26;
select database();
```

```
# Find out average number of copies of each book
select book_id, avg(no_of_copies) as Average_No_of_Copies_Per_Book
from Book_Copies
group by book_id;
```

```
+-----+-----+
| book_id | Average_No_of_Copies_Per_Book |
+-----+-----+
|      1 |              7.5000 |
|      2 |              3.5000 |
|      3 |              7.0000 |
|      4 |              3.0000 |
|      5 |              1.0000 |
|      6 |             11.5000 |
+-----+-----+
6 rows in set (0.0010 sec)
```

Display the details of books along with average number of copies of each book

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id, author_name, title;
```

```
+-----+-----+-----+-----+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
|      1 | Navathe    | DBMS  |              7.5000 |
|      2 | Navathe    | ADBMS |              3.5000 |
|      3 | Tanenbaum  | CN    |              7.0000 |
|      4 | Edward Angel | CG    |              3.0000 |
|      5 | Galvin     | OS    |              1.0000 |
|      6 | K. P. Adhiya | CO    |             11.5000 |
+-----+-----+-----+-----+
6 rows in set (0.0010 sec)
```

Find out Authors whose average no of book copies in library are more than 3 and display the name of author on top having maximum average number of book copies.

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id,author_name, title
having Average_No_of_Copies_Per_Book >3
order by Average_No_of_Copies_Per_Book desc;
```

```
+-----+-----+-----+-----+
-+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
-+
|      6 | K. P. Adhiya | CO      | 11.5000 |
|      1 | Navathe      | DBMS    | 7.5000  |
|      3 | Tanenbaum    | CN      | 7.0000  |
|      2 | Navathe      | ADBMS   | 3.5000  |
+-----+-----+-----+-----+
-+
4 rows in set (0.0008 sec)
```

```
# Find out Author's name who wrote maximum books
select author_name, count(author_name) as Number_of_Books_Written
from Book_Authors
group by author_name
order by Number_of_Books_Written desc
limit 1;
```

```
+-----+-----+
| author_name | Number_of_Books_Written |
+-----+-----+
| Navathe     | 2 |
+-----+-----+
1 row in set (0.0009 sec)
```

```
# Find out total number of Publishers
select count(*) as Total_No_of_Publishers
from Publisher;
```

```
+-----+
| Total_No_of_Publishers |
+-----+
| 11 |
+-----+
1 row in set (0.0007 sec)
```

Proper Query: - Find out the book and its corresponding branch having maximum number of copies

```
select *
from book_copies
where no_of_copies =
      (select max(no_of_copies)
       from book_copies);
```

```
+-----+-----+-----+
| book_id | branch_id | no_of_copies |
+-----+-----+-----+
|        6 |         15 |           13 |
+-----+-----+-----+
1 row in set (0.0009 sec)
```

```
select *
from book_copies
where no_of_copies =
      (select min(no_of_copies)
       from book_copies);
```

```
+-----+-----+-----+
| book_id | branch_id | no_of_copies |
+-----+-----+-----+
|        5 |         10 |           1 |
+-----+-----+-----+
1 row in set (0.0007 sec)
```

Display total number of copies of all books

```
select sum(no_of_copies) as Total_Copies_of_All_Books
from Book_Copies;
```

```
+-----+
| Total_Copies_of_All_Books |
+-----+
|                          56 |
+-----+
1 row in set (0.0006 sec)
```

List title of books and publication year, published between 2015 to 2022. Display recent publishing year on top.

```
select title, pub_year
from Book
where pub_year between '2015' and '2022'
order by pub_year desc;
```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                  | 2022     |
| DBMS for Distributed System | 2022     |
| DBMS                  | 2017     |
| ADBMS                 | 2016     |
| CN                    | 2016     |
| OS                    | 2016     |
| CG                    | 2015     |
+-----+-----+

```

7 rows in set (0.0005 sec)

List title of books and publication year, published between 2017 to 2022.

```

select title, pub_year
from Book
where pub_year between '2017' and '2022';

```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                  | 2017     |
| DBMS                  | 2022     |
| DBMS for Distributed System | 2022     |
+-----+-----+

```

3 rows in set (0.0006 sec)

Find out average number of copies of each book

```

select book_id, avg(no_of_copies) as Average_No_of_Copies_Per_Book
from Book_Copies
group by book_id;

```

```

+-----+-----+
| book_id | Average_No_of_Copies_Per_Book |
+-----+-----+
| 1       | 7.5000 |
| 2       | 3.5000 |
| 3       | 7.0000 |
| 4       | 3.0000 |
| 5       | 1.0000 |
| 6       | 11.5000 |
+-----+-----+

```

6 rows in set (0.0010 sec)

Find out Authors whose average no of book copies in library are more than 3 and display the name of author on top having maximum average number of book copies.

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id,author_name, title
having Average_No_of_Copies_Per_Book >3
order by Average_No_of_Copies_Per_Book desc;
```

```
+-----+-----+-----+-----+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
|      6 | K. P. Adhiya | CO    | 11.5000 |
|      1 | Navathe      | DBMS  | 7.5000  |
|      3 | Tanenbaum    | CN    | 7.0000  |
|      2 | Navathe      | ADBMS | 3.5000  |
+-----+-----+-----+-----+
4 rows in set (0.0008 sec)
```

Display the details of publishers from Pune and Jalgaon.

```
select *
from publisher
where address = 'Pune' or address = 'Jalgaon';
```

```
select *
from publisher
where address in ('Pune','Jalgaon');
```

```
+-----+-----+-----+
| name                | address | phone    |
+-----+-----+-----+
| Nirali Prakashan    | Pune    | 2025512336 |
| Prakash Publication | Jalgaon | 9272514482 |
| Technical Publications | Pune    | 9763719464 |
+-----+-----+-----+
3 rows in set (0.0045 sec)
```

#Display the details of publishers other than Pune and Jalgaon.

```
select *
from publisher
where address not in ('Pune','Jalgaon');
```

name	address	phone
Galgotia Publications	New Delhi	1123263334
Grupo Planeta	Bangalore	7756120238
Hachette Livre	Chennai	8970862340
Mcgraw-Hill	Bangalore	9989076587
Pearson	New Delhi	9889076565
Random House	Hyderabad	7455679345
S Chand	Noida	7291975264
Wiley India	New Delhi	1143630000

8 rows in set (0.0006 sec)

Experiment No. 6

#To use various MySQL Commands/Queries such as - Set
#Operations e.g. UNION, UNION ALL, Set Membership
#Operations e.g. IN, NOT IN, BETWEEN, Set Comparison
#Operations e.g. SOME (or ANY), ALL.

```
use Library_DB_SE_2025_26;  
select database();
```

Display Publishers name from New Delhi and Pune cities, using UNION
operator

```
select name, address  
from Publisher  
where address = 'New Delhi'  
union  
select name, address  
from Publisher  
where address = 'Pune';
```

```
select name, address  
from Publisher  
where address = 'New Delhi'  
union all  
select name, address  
from Publisher  
where address = 'Pune';
```

```
+-----+-----+  
| name                | address  |  
+-----+-----+  
| Galgotia Publications | New Delhi |  
| Pearson              | New Delhi |  
| Wiley India          | New Delhi |  
| Nirali Prakashan     | Pune     |  
| Technical Publications | Pune     |  
+-----+-----+  
5 rows in set (0.0009 sec)
```



```
# Display the identification number of books and their corresponding
author name, using Book table and Book_Authors table
```

```
# Use of join on
```

```
select Book.book_id, Book_authors.book_id, author_name
from Book join Book_Authors
on Book.book_id = Book_Authors.book_id;
```

```
# Use of inner join on
```

```
select Book.book_id, Book_authors.book_id, author_name
from Book inner join Book_Authors
on Book.book_id = Book_Authors.book_id;
```

```
#Use of natural join
```

```
select Book.book_id, Book_authors.book_id, author_name
from Book natural join Book_Authors;
```

```
+-----+-----+-----+
| book_id | book_id | author_name |
+-----+-----+-----+
|      7 |      7 | Navin Kumar |
|      4 |      4 | Edward Angel |
|      1 |      1 | Navathe     |
|      2 |      2 | Navathe     |
|      6 |      6 | K. P. Adhiya |
|      3 |      3 | Tanenbaum   |
|      5 |      5 | Galvin      |
|      8 |      8 | A. A. Puntambekar |
+-----+-----+-----+
8 rows in set (0.0005 sec)
```

```
# Display the details of books along with average number of copies
of each book
```

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id, author_name, title;
```

```
+-----+-----+-----+-----+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
|      1 | Navathe     | DBMS  | 7.5000 |
|      2 | Navathe     | ADBMS | 3.5000 |
|      3 | Tanenbaum   | CN     | 7.0000 |
|      4 | Edward Angel | CG     | 3.0000 |
|      5 | Galvin      | OS     | 1.0000 |
|      6 | K. P. Adhiya | CO     | 11.5000 |
+-----+-----+-----+-----+
6 rows in set (0.0006 sec)
```

```
# Display the details of publishers from Pune and Jalgaon.
```

```
select *
from publisher
where address = 'Pune' or address = 'Jalgaon';
```

```
select *
from publisher
where address in ('Pune','Jalgaon');
```

```
+-----+-----+-----+
| name                | address | phone      |
+-----+-----+-----+
| Nirali Prakashan    | Pune    | 2025512336 |
| Prakash Publication | Jalgaon | 9272514482 |
| Technical Publications | Pune    | 9763719464 |
+-----+-----+-----+
3 rows in set (0.0045 sec)
```

```
#Display the details of publishers other than Pune and Jalgaon.
```

```
select *
from publisher
where address not in ('Pune','Jalgaon');
```

```
+-----+-----+-----+
| name                | address | phone      |
+-----+-----+-----+
| Galgotia Publications | New Delhi | 1123263334 |
| Grupo Planeta         | Bangalore | 7756120238 |
| Hachette Livre        | Chennai  | 8970862340 |
| Mcgraw-Hill           | Bangalore | 9989076587 |
| Pearson               | New Delhi | 9889076565 |
| Random House          | Hyderabad | 7455679345 |
| S Chand               | Noida    | 7291975264 |
| Wiley India           | New Delhi | 1143630000 |
+-----+-----+-----+
8 rows in set (0.0006 sec)
```

```
# List title of books and publication year, published between 2015
to 2022. Display recent publishing year on top.
```

```
select title, pub_year
from Book
where pub_year between '2015' and '2022'
order by pub_year desc;
```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                 | 2022     |
| DBMS for Distributed System | 2022     |
| DBMS                 | 2017     |
| ADBMS                | 2016     |
| CN                   | 2016     |
| OS                   | 2016     |
| CG                   | 2015     |
+-----+-----+
7 rows in set (0.0005 sec)

```

List title of books and publication year, published between 2017 to 2022.

```

select title, pub_year
from Book
where pub_year between '2017' and '2022';

```

```

+-----+-----+
| title                | pub_year |
+-----+-----+
| DBMS                 | 2017     |
| DBMS                 | 2022     |
| DBMS for Distributed System | 2022     |
+-----+-----+
3 rows in set (0.0006 sec)

```

```

select title, pub_year
from Book
where pub_year not between '2015' and '2022'
order by pub_year desc;

```

```

+-----+-----+
| title | pub_year |
+-----+-----+
| CO    | 2008     |
| CO    | 2005     |
+-----+-----+
2 rows in set (0.0004 sec)

```

Use of Set comparison operator - SOME/ANY operator

```
select book_id, branch_id
from book_copies
where no_of_copies >any
      (select no_of_copies
       from book_copies
       where book_id = 1);
```

```
+-----+-----+
| book_id | branch_id |
+-----+-----+
|      1 |      10 |
|      3 |      14 |
|      6 |      15 |
|      6 |      16 |
+-----+-----+
4 rows in set (0.0007 sec)
```

Use of Set comparison operator - ALL operator

```
select book_id, branch_id
from book_copies
where no_of_copies >all
      (select no_of_copies
       from book_copies
       where book_id = 1);
```

```
+-----+-----+
| book_id | branch_id |
+-----+-----+
|      6 |      15 |
+-----+-----+
1 row in set (0.0006 sec)
```

Experiment No. 7

To use various MySQL Commands/Queries related to Join
operations, such as - INNER JOIN, LEFT JOIN, RIGHT JOIN,
Full Join using UNION.

```
use Library_DB_SE_2025_26;  
select database();
```

Display the identification number of books and their corresponding
author name, using Book table and Book_Authors table

Use of join on

```
select Book.book_id, Book_authors.book_id, author_name  
from Book join Book_Authors
```

```
on Book.book_id = Book_Authors.book_id;
```

Use of inner join on

```
select Book.book_id, Book_authors.book_id, author_name  
from Book inner join Book_Authors
```

```
on Book.book_id = Book_Authors.book_id;
```

#Use of natural join

```
select Book.book_id, Book_authors.book_id, author_name  
from Book natural join Book_Authors;
```

```
+-----+-----+-----+  
| book_id | book_id | author_name |  
+-----+-----+-----+  
|      7 |      7 | Navin Kumar |  
|      4 |      4 | Edward Angel |  
|      1 |      1 | Navathe     |  
|      2 |      2 | Navathe     |  
|      6 |      6 | K. P. Adhiya |  
|      3 |      3 | Tanenbaum   |  
|      5 |      5 | Galvin      |  
|      8 |      8 | A. A. Puntambekar |  
+-----+-----+-----+
```

8 rows in set (0.0005 sec)

Display the details of books along with average number of copies
of each book

```
select Book_Copies.book_id, author_name, title, avg(no_of_copies)  
as Average_No_of_Copies_Per_Book  
from Book_Copies join Book_Authors join Book  
on Book_Copies.book_id = Book_Authors.book_id and Book.book_id =  
Book_Authors.book_id  
group by book_id, author_name, title;
```

book_id	author_name	title	Average_No_of_Copies_Per_Book
1	Navathe	DBMS	7.5000
2	Navathe	ADBMS	3.5000
3	Tanenbaum	CN	7.0000
4	Edward Angel	CG	3.0000
5	Galvin	OS	1.0000
6	K. P. Adhiya	CO	11.5000

6 rows in set (0.0006 sec)

```
# List title of those books who don't have any author
select title, author_name
from book left join book_authors
on book.book_id = book_authors.book_id
where book_authors.author_name is null;
```

```
select title, author_name
from book_authors right join book
on book.book_id = book_authors.book_id
where book_authors.author_name is null;
```

title	author_name
DBMS for Distributed System	NULL

1 row in set (0.0004 sec)

```
# Use of full join concept
select title
from book left join book_authors
on book.book_id = book_authors.book_id
where book_authors.author_name is null
union
select title
from book_authors right join book
on book.book_id = book_authors.book_id
where book_authors.author_name is null;
```

title
DBMS for Distributed System

1 row in set (0.0009 sec)

```

select title
from book left join book_authors
on book.book_id = book_authors.book_id
where book_authors.author_name is null
union all
select title
from book_authors right join book
on book.book_id = book_authors.book_id
where book_authors.author_name is null;

```

```

+-----+
| title |
+-----+
| DBMS for Distributed System |
| DBMS for Distributed System |
+-----+
2 rows in set (0.0006 sec)

```

```

# Display the details of books along with average number of copies
# of each book
# Use of equi-join

```

```

select Book_Copies.book_id, author_name, title, avg(no_of_copies)
as Average_No_of_Copies_Per_Book
from Book_Copies, Book_Authors, Book
where Book_Copies.book_id = Book_Authors.book_id and Book.book_id
= Book_Authors.book_id
group by book_id, author_name, title;

```

```

+-----+-----+-----+-----+
| book_id | author_name | title | Average_No_of_Copies_Per_Book |
+-----+-----+-----+-----+
| 1 | Navathe | DBMS | 7.5000 |
| 2 | Navathe | ADBMS | 3.5000 |
| 3 | Tanenbaum | CN | 7.0000 |
| 4 | Edward Angel | CG | 3.0000 |
| 5 | Galvin | OS | 1.0000 |
| 6 | K. P. Adhiya | CO | 11.5000 |
+-----+-----+-----+-----+
6 rows in set (0.0008 sec)

```

```

# Display the identification number of books and their corresponding
author name, using Book table and Book_Authors table
# Use of join on
select Book.book_id, Book_authors.book_id, author_name
from Book join Book_Authors
on Book.book_id = Book_Authors.book_id;

```

```
# Use of inner join on
select Book.book_id, Book_authors.book_id, author_name
from Book inner join Book_Authors
on Book.book_id = Book_Authors.book_id;
#Use of natural join
select Book.book_id, Book_authors.book_id, author_name
from Book natural join Book_Authors;
```

```
+-----+-----+-----+
| book_id | book_id | author_name |
+-----+-----+-----+
|      7 |      7 | Navin Kumar |
|      4 |      4 | Edward Angel |
|      1 |      1 | Navathe      |
|      2 |      2 | Navathe      |
|      6 |      6 | K. P. Adhiya |
|      3 |      3 | Tanenbaum    |
|      5 |      5 | Galvin       |
|      8 |      8 | A. A. Puntambekar |
+-----+-----+-----+
8 rows in set (0.0005 sec)
```


Experiment No. 8

To create a trigger and write a program to demonstrate
its use in MySQL.

```
use Library_DB_SE_2025_26;  
select database();
```

```
# Trigger  
# If number of copies of any book purchased for any branch is more  
# than 10, then as a discount, branch receives one more additional  
# copy. For more than 20, additional 2 copies and so on. (i.e. one  
# more additional copy per 10 books)
```

```
delimiter $$  
create trigger trigger_befor_insert_Lib_DB before insert  
on book_copies for each row  
begin  
    if new.no_of_copies>10  
    then  
        set new.no_of_copies = new.no_of_copies +  
                                floor(new.no_of_copies * 0.10);  
    end if;  
end $$  
delimiter ;
```

```
show triggers;
```

```
select * from book_copies;
```

```
+-----+-----+-----+  
| book_id | branch_id | no_of_copies |  
+-----+-----+-----+  
|      1 |      10 |           10 |  
|      1 |      11 |            5 |  
|      2 |      12 |            2 |  
|      2 |      13 |            5 |  
|      3 |      14 |            7 |  
|      4 |      11 |            3 |  
|      5 |      10 |            1 |  
|      6 |      15 |           13 |  
|      6 |      16 |           10 |  
+-----+-----+-----+  
9 rows in set (0.0009 sec)
```

```

insert into book_copies values(3, 15, 11);
select * from book_copies;
insert into book_copies values(3, 16, 17);
select * from book_copies;
insert into book_copies values(3, 13, 21);
select * from book_copies;

```

book_id	branch_id	no_of_copies
1	10	10
1	11	5
2	12	2
2	13	5
3	13	23
3	14	7
3	15	12
3	16	18
4	11	3
5	10	1
6	15	13
6	16	10

```

+-----+-----+-----+
12 rows in set (0.0005 sec)

```

```

delete from book_copies
where (book_id=3 and branch_id=15)
or (book_id=3 and branch_id=16)
or (book_id=3 and branch_id=13);

```

```

select * from book_copies;

```

```

drop trigger trigger_befor_insert_Lib_DB;

```

```

show triggers;

```

Experiment No. 9

To create a View and to perform various operations on a
View in MySQL.

```
use Library_DB_SE_2025_26;  
select database();
```

View

To create a view for students - To display his/her card number,
book id & book title against his/her name, branch id & branch name
from where book is issued and due date.

```
create view View_for_Student_Lib_DB as  
select card_no, book_loans.book_id, title, book_loans.branch_id,  
                                             branch_name, due_date  
from book_loans, book, library_branch  
where book_loans.branch_id = library_branch.branch_id and  
book_loans.book_id = book.book_id;
```

```
show full tables;
```

```
+-----+-----+  
| Tables_in_library_db_se_2025_26 | Table_type |  
+-----+-----+  
| book                               | BASE TABLE |  
| book_authors                       | BASE TABLE |  
| book_copies                        | BASE TABLE |  
| book_loans                         | BASE TABLE |  
| borrower                           | BASE TABLE |  
| library_branch                     | BASE TABLE |  
| publisher                          | BASE TABLE |  
| view_for_student_lib_db            | VIEW         |  
+-----+-----+  
8 rows in set (0.0012 sec)
```

```
select * from View_for_Student_Lib_DB;
```

card_no	book_id	title	branch_id	branch_name	due_date
101	1	DBMS	10	RR Nagar	2017-06-17
104	1	DBMS	11	RNSIT	2017-05-12
101	2	ADBMS	13	NITTE	2017-04-21
101	3	CN	14	Manipal	2017-07-17
101	4	CG	11	RNSIT	2017-07-15

5 rows in set (0.0007 sec)

```
select *  
from View_for_Student_Lib_DB  
where card_no=104;
```

card_no	book_id	title	branch_id	branch_name	due_date
104	1	DBMS	11	RNSIT	2017-05-12

1 row in set (0.0010 sec)

```
# Update a View  
update View_for_Student_Lib_DB  
set due_date = '2017-07-12'  
where card_no = 104;
```

```
select *  
from View_for_Student_Lib_DB  
where card_no=104;  
select * from book_loans  
where card_no=104;
```

card_no	book_id	title	branch_id	branch_name	due_date
104	1	DBMS	11	RNSIT	2017-07-12

1 row in set (0.0007 sec)

```
update View_for_Student_Lib_DB  
set due_date = '2017-05-12'  
where card_no = 104;
```

```
select *  
from View_for_Student_Lib_DB  
where card_no=104;  
select * from book_loans  
where card_no=104;
```

```
# delete a View
drop view View_for_Student_Lib_DB;
show full tables;
```

```
+-----+-----+
| Tables_in_library_db_se_2025_26 | Table_type |
+-----+-----+
| book                               | BASE TABLE |
| book_authors                       | BASE TABLE |
| book_copies                        | BASE TABLE |
| book_loans                         | BASE TABLE |
| borrower                           | BASE TABLE |
| library_branch                     | BASE TABLE |
| publisher                           | BASE TABLE |
+-----+-----+
7 rows in set (0.0012 sec)
```

Experiment No. 10

To create a stored Function and its use in MySQL.

```
use Library_DB_SE_2025_26;
select database();
```

Function

The function is used to display number of publishers from a
particular city

```
delimiter $$
create function publisher_count(publisher_address_input
varchar(30))
    returns integer
    deterministic
    begin
        declare p_count integer;
        select count(*) into p_count
        from publisher
        where publisher.address = publisher_address_input;
        return p_count;
    end
$$
delimiter ;
```

```
show function status;
```

```
select publisher_count ('New Delhi') as Publisher_Count;
+-----+
| Publisher_Count |
+-----+
|                3 |
+-----+
1 row in set (0.0008 sec)
```

```
select publisher_count ('Mumbai') as Publisher_Count;
+-----+
| Publisher_Count |
+-----+
|                0 |
+-----+
1 row in set (0.0006 sec)
```

```
select publisher_count ('Bangalore') as Publisher_Count;
```

```
+-----+
```

```
| Publisher_Count |
```

```
+-----+
```

```
|                2 |
```

```
+-----+
```

```
1 row in set (0.0005 sec)
```

```
drop function publisher_count;
```

```
show function status;
```

Experiment No. 11

To create a stored Procedure and its use in MySQL, and
demonstrate its use using CALL keyword.

```
use Library_DB_SE_2025_26;  
select database();
```

Procedure

To define the procedure that returns the count of publishers

```
delimiter $$  
create procedure publisher_count_procedure(in  
publisher_address_input varchar(30), out p_count int)  
begin  
    select count(*) into p_count  
    from publisher  
    where publisher.address = publisher_address_input;  
end $$  
delimiter ;
```

```
show procedure status;
```

Now call the procedure

```
call publisher_count_procedure ('New Delhi', @p_count);  
select @p_count as Publisher_Count;
```

```
+-----+  
| Publisher_Count |  
+-----+  
|                3 |  
+-----+  
1 row in set (0.0003 sec)
```

```
call publisher_count_procedure ('Mumbai', @p_count);  
select @p_count as Publisher_Count;
```

```
+-----+  
| Publisher_Count |  
+-----+  
|                0 |  
+-----+  
1 row in set (0.0004 sec)
```



```
call publisher_count_procedure ('Bangalore', @pub_count);
select @pub_count as Publisher_Count;
```

```
+-----+
| Publisher_Count |
+-----+
|                2 |
+-----+
1 row in set (0.0004 sec)
```

```
drop procedure publisher_count_procedure;
show procedure status;
```